Numerical Simulation for Evolution of Water Meniscus in Unsaturated Clay using Peridynamics

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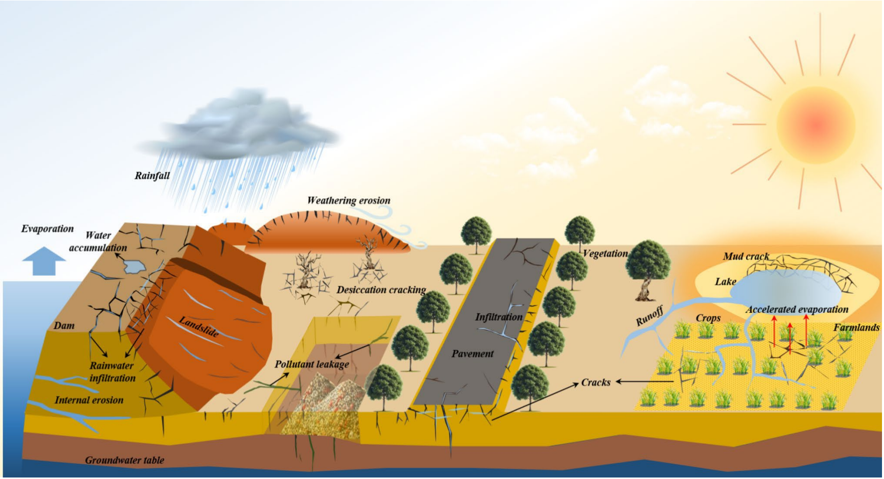
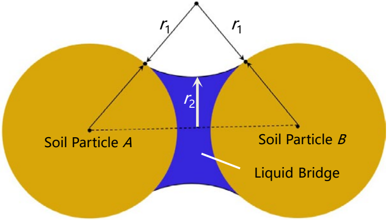
Contents

Motivation

Desiccation cracking of clay is intensified by global warming and extreme drought.



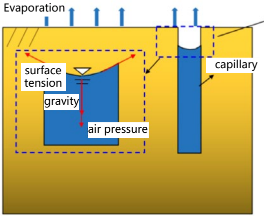
The single liquid bridge model of unsaturated clay

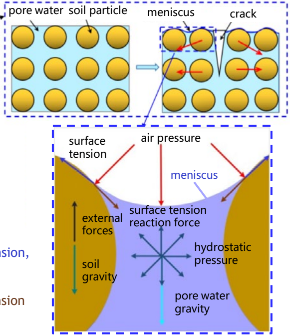


The matrix suction of unsaturated soil is defined by Young-Laplace Equation, related to the morphology curved liquid surface

$$s = \gamma_s \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

Micro-scale meniscus evolution controls capillary suction → governs macro-scale shrinkage and cracking!



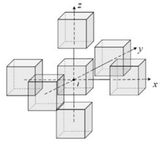


Force on pore water:
Air pressure, hydrostatic pressure, surface tension, gravity
Force on soil:
Air pressure, hydrostatic pressure, surface tension reaction force, gravity, other external forces

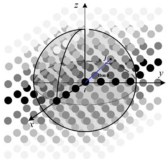
Limitations of existing numerical methods

Research Methods	Model	Features	Limitation
DDA	General Model	The soil mass is discretized into irregular polygonal particles	The solid-liquid contact angle and radius of curvature of a meniscus are constant, making it impossible to obtain information about its dynamic evolution
	Two-dimensional extend model	Determining the morphology of the meniscus based on the water content per unit volume of soil	
DEM	General model	The effect of surface tension is replaced by adhesive force	Lack of description of menisci
	Isolated capillary model	Suitable for two-dimensional soils with low saturation	Introducing numerous parameters to describe the morphology of the meniscus may introduce significant calculation errors and fails to capture the dynamic evolution process
	Cableway liquid bridge model	Suitable for highly saturated two-dimensional soils	
	Complex fuid cluster model	Suitable for highly saturated three-dimensional soils	
CFD-DEM	One-way fluid-structure interaction model	Using CFD method to simulate water pressure, using DEM method to simulate soil particle movement, considering unsaturated soil matrix suction using constant adhesion force	The flow field is not affected by the movement of soil particles and cannot fully reflect the fluid-structure interaction under surface tension
MD	General model	Nanoscale liquid bridge model	Modeling and calculation based on molecular force fields Lack of description of surface tension

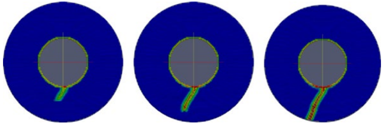
PD modeling in soil deformation



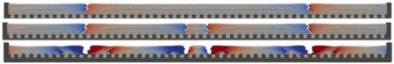
Local model of continuum mechanics



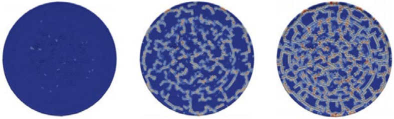
Non-local model of peridynamics (Silling, 2000)



(Jabakhani, 2013)

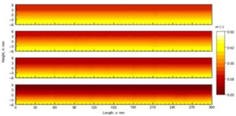


Evolution of cracking in soil strip

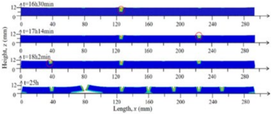


Evolution of crazing in soil plate

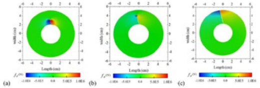
(Menon and Song, 2019)



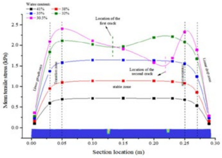
Moisture transport in soil strip



Cracking of soil strip



Evolution of stress in soil ring cracking



Distribution of stress in soil strip cracking

(Yan et al., 2023; Liu et al., 2023)

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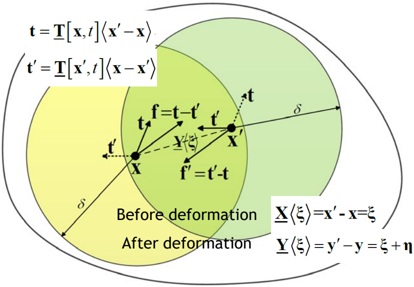
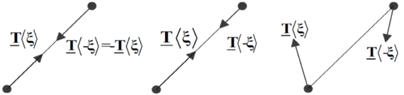
PD framework and PDDO

Interaction between material points in PD modeling framework



(Silling, 2008)



Bond-based | Ordinary state-based | Unordinary state-based

Peridynamic differential operators

$$\frac{\partial^{n_1+n_2+\cdots n_M} f(\mathbf{x})}{\partial x_1^{p_1} \partial x_2^{p_2} \cdots \partial x_M^{p_M}} = \int_{H_\mathbf{x}} f(\mathbf{x} + \boldsymbol{\xi}) g_N^{p_1 p_2 \cdots p_N}(\boldsymbol{\xi}) \mathrm{d}V_{\mathbf{x}'}$$

(Madenci et al., 2016)

Governing equations of free surface flow $\xrightarrow{\text{PDDO}}$ Non-local form expression

Lagrangian description

$$\frac{\partial \rho}{\partial t} = -\rho \nabla \cdot \mathbf{v}$$

$$\rho \frac{\partial \mathbf{v}}{\partial t} = -\nabla P + \mathbf{F_V} + \mathbf{F_B} + \mathbf{F_S}$$

where

$$P = \frac{\rho_0 c_f^2}{\gamma} \left(\left(\frac{\rho}{\rho_0} \right)^{\gamma} - 1 \right) + P_0$$

$$\mathbf{F_V} = \mu(\Delta \mathbf{v})$$

$$\mathbf{F_B} = 0$$

$$\mathbf{F_S} = \beta \kappa \mathbf{n} \nabla s$$

PD integral equations

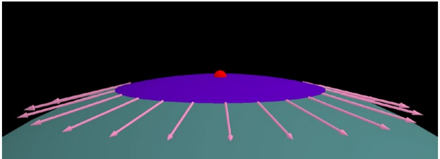
$$\frac{\partial \rho(\mathbf{x})}{\partial t} = -\rho(\mathbf{x}) \int_{H_\mathbf{x}} \mathbf{g}_1(\boldsymbol{\xi}) \cdot \big(\mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})\big) \mathrm{d}V_{\mathbf{x}'}$$

$$\rho \frac{\partial \mathbf{v}}{\partial t} = -\int_{H_\mathbf{x}} \big(P(\mathbf{x}') - P(\mathbf{x})\big) \mathbf{g}_1(\boldsymbol{\xi}) dV_{\mathbf{x}'} + \int_{H_\mathbf{x}} \mu(\mathbf{x}) \big(\mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})\big) \mathrm{tr}\big(\mathbf{g}_2(\boldsymbol{\xi})\big) dV_{\mathbf{x}'} - \beta \int_{H_\mathbf{x}} \kappa (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\boldsymbol{\xi}) \mathrm{d}V_{\mathbf{x}'}$$

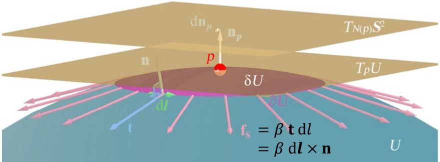
where

$$\kappa = \frac{\int_{H_\mathbf{x}} (\mathbf{n}(\mathbf{x}') - \mathbf{n}(\mathbf{x})) \cdot \mathbf{g}_1(\xi) \mathrm{d}V_{\mathbf{x}'}}{\int_{H_\mathbf{x}} w_0(\|\boldsymbol{\xi}\|) \mathrm{d}V_{\mathbf{x}'} / \int_{H_\mathbf{x}} w_0(\|\boldsymbol{\xi}\|) \mathrm{d}V_{\mathbf{x}'}}$$

Calculation of surface tension



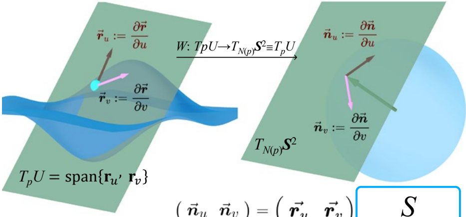
A smooth and singularity-free surface that satisfies: Gauss' consistency condition and Mainardi-Codazzi-Peterson consistency condition



$$\mathbf{f_s} = \beta\,\mathbf{t}\,\mathrm{d}l = \beta\,\mathrm{d}\boldsymbol{l} \times \mathbf{n}$$

$$\delta\mathbf{F_S} = \oint_{\partial U} \beta \mathrm{d}\boldsymbol{l} \times \mathbf{n} = \oint_{\partial U} \beta \mathrm{d}\boldsymbol{l} \times \mathbf{n_p} + \boxed{\oint_{\partial U} \beta \mathrm{d}\boldsymbol{l} \times \mathrm{d}\mathbf{n_p}} = -\mathbf{n_p} \oint_{\partial U} (\beta \mathrm{d}\mathbf{n_p}) \cdot (\mathbf{t}\,\mathrm{d}l)$$

$$\vec{r}(u,v) = \begin{pmatrix} x(u,v) \\ y(u,v) \\ z(u,v) \end{pmatrix} \xrightarrow{N:\, U \to S^2 \subset \mathbf{R}^3} \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|} \boxed{\vec{n}(u,v)} = \begin{pmatrix} n_1(u,v) \\ n_2(u,v) \\ n_3(u,v) \end{pmatrix}$$

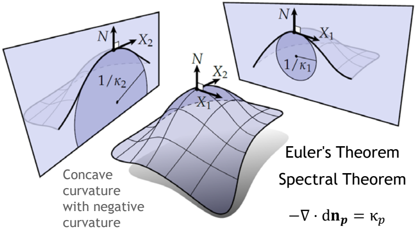


$$\begin{pmatrix} \vec{n}_u & \vec{n}_v \end{pmatrix} = \begin{pmatrix} \vec{r}_u & \vec{r}_v \end{pmatrix} \boxed{S}$$

Shape Matrix

Gauss theorem $\Rightarrow \mathbf{n_p} \oint_{\partial U} (\beta \mathrm{d}\mathbf{n_p}) \cdot (\mathbf{t}\,\mathrm{d}l) = \beta(\nabla \cdot \mathrm{d}\mathbf{n_p})\delta U\, \mathbf{n_p}$

$$\mathbf{F_S} = \lim_{\delta U \to 0} \frac{\delta \mathbf{F_S}}{\delta U} = \beta(-\nabla \cdot \mathrm{d}\mathbf{n_p})\, \mathbf{n_p}$$



Euler's Theorem

Spectral Theorem

$$-\nabla \cdot \mathrm{d}\mathbf{n_p} = \kappa_p$$

Calculation process

- Continuity Equation

$$\rho_i^{n+1} = \rho_i^n - \rho_i^n \Delta t \sum_{j=1}^{N_i} \left[\left(\mathbf{v}_j^n - \mathbf{v}_i^n \right) \cdot \mathbf{g}_1 \left(\boldsymbol{\xi}_{ij}^n \right) \right] V_j$$

$$\rho_i^{n+1} = \rho_i^n - \frac{2\Delta x^2}{\pi \delta^2} \rho_i^n \Delta t \sum_{j=1}^{N_i} \left(\mathbf{v}_j^n - \mathbf{v}_i^n \right) \frac{[\xi_{ij1} \quad \xi_{ij2}]^{\mathrm{T}}}{\left\| \xi_{ij1}^2 + \xi_{ij2}^2 \right\|}$$

For two-dimensional problems, the relative value form of the near-field dynamic differential operator is adopted.

- Momentum equation

$$\mathbf{a}_i^{n+1} = \frac{1}{\rho_i^{n+1}} \left(\begin{array}{c} -\sum_{j=1}^{N_i} \left(P_j^n - P_i^n \right) \mathbf{g}_1 \left(\boldsymbol{\xi}_{ij}^n \right) V_j + \sum_{j=1}^{N_i} \mu_i \left(\mathbf{v}_j^n - \mathbf{v}_i^n \right) \mathrm{tr} \left(\mathbf{g}_2 \left(\boldsymbol{\xi}_{ij}^n \right) \right) V_j \\ -\beta \dfrac{\left(\sum_{j=1}^{N_i} \min(\mathbb{N}_i, \mathbb{N}_j) \left(\mathbf{n}_j^n - \mathbf{n}_i^n \right) \cdot \mathbf{g}_1 \left(\boldsymbol{\xi}_{ij}^n \right) V_j \right)}{\sum_{j=1}^{N_i} \min(\mathbb{N}_i, \mathbb{N}_j) w_0 \left(\left\| \boldsymbol{\xi}_{ij}^n \right\| \right) V_j / \sum_{j=1}^{N_i} w_0 \left(\left\| \boldsymbol{\xi}_{ij}^n \right\| \right) V_j} \left(\sum_{j=1}^{N_i} \left(s(\mathbf{x}') - s(\mathbf{x}) \right) \mathbf{g}_1 \left(\boldsymbol{\xi}_{ij}^n \right) V_j \right) \end{array} \right) + \frac{1}{\rho_i^{n+1} \mathbf{F}^B}$$

$$\mathbf{a}_i^{n+1} = \frac{2\Delta x^2}{\pi \delta^2 \rho_i^{n+1}} \left(\begin{array}{c} -\sum_{j=1}^{N_i} \left(P_j^n - P_i^n \right) \dfrac{[\xi_{ij1} \quad \xi_{ij2}]^{\mathrm{T}}}{\left\| \xi_{ij1}^2 + \xi_{ij2}^2 \right\|} + \dfrac{3\mu_i}{\delta} \sum_{j=1}^{N_i} \dfrac{\left(\mathbf{v}_j^n - \mathbf{v}_i^n \right)}{\sqrt{\left\| \xi_{ij1}^2 + \xi_{ij2}^2 \right\|}} \\ -\dfrac{\left(\sum_{j=1}^{N_i} \min(\mathbb{N}_i, \mathbb{N}_j) \left(\mathbf{n}_j^n - \mathbf{n}_i^n \right) \dfrac{[\xi_{ij1} \quad \xi_{ij2}]^{\mathrm{T}}}{\left\| \xi_{ij1}^2 + \xi_{ij2}^2 \right\|} \right) \left(\sum_{j=1}^{N_i} \left(s(\mathbf{x}') - s(\mathbf{x}) \right) \dfrac{[\xi_{ij1} \quad \xi_{ij2}]^{\mathrm{T}}}{\left\| \xi_{ij1}^2 + \xi_{ij2}^2 \right\|} \right)}{\dfrac{\pi \delta^2}{2\beta \Delta x^2} \left(\sum_{j=1}^{N_i} \min(\mathbb{N}_i, \mathbb{N}_j) w_0 \left(\left\| \boldsymbol{\xi}_{ij}^n \right\| \right) V_j / \sum_{j=1}^{N_i} w_0 \left(\left\| \boldsymbol{\xi}_{ij}^n \right\| \right) V_j \right)} \end{array} \right) + \frac{1}{\rho_i^{n+1} \mathbf{F}^B}$$

- Time difference scheme

Second-order Verlet-Velocity algorithm

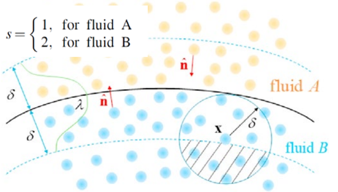
$$\begin{cases} \mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \dfrac{1}{2} \left(\mathbf{a}_i^{n+1} + \mathbf{a}_i^n \right) \Delta t \\ \mathbf{u}_i^{n+1} = \mathbf{u}_i^n + \mathbf{v}_i^n \Delta t + \dfrac{1}{2} \mathbf{a}_i^n \Delta t^2 \\ \mathbf{x}_i^{n+1} = \mathbf{x}_i^0 + \mathbf{u}_i^{n+1} \qquad \mathbf{u}_i^0 \end{cases}$$

Courant-Friedrichs-Levy stability condition

$$\Delta t < \min \left(\frac{h_{\mathrm{smooth}}}{|\mathbf{v}_{\max}| + c_B}, \frac{1}{8} \frac{{h_{\mathrm{smooth}}}^2}{\max(\mu/\rho_0)}, \frac{1}{4} \sqrt{\frac{h_{\mathrm{smooth}}}{g}}, \frac{1}{4} \sqrt{\frac{\min(\rho_0) h^3}{2\pi\beta}} \right)$$

The non-local form of the surface tension formula

$$s = \begin{cases} 1, & \text{for fluid A} \\ 2, & \text{for fluid B} \end{cases}$$



$$\left(s(x') - s(\mathbf{x}) \right) = \mathbb{C}(x', \mathbf{x}) = \begin{cases} \dfrac{2\rho_{\mathbf{x},0}}{\rho_{\mathbf{x},0} + \rho_{x',0}}, & \text{if } \mathbf{x} \text{ and } x' \text{ belong to different phases} \\ 0, & \text{if } \mathbf{x} \text{ and } x' \text{ belong to the same phase} \end{cases}$$

Introduce the truncation function to avoid the unit normal vector field having incorrect directions

$$\mathbf{n}(\mathbf{x}) = \begin{cases} \dfrac{\int_{H_{\mathbf{x}}} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) \mathrm{d}V_{\mathbf{x}'}}{\left\| \int_{H_{\mathbf{x}}} (s(\mathbf{x}') - s(\mathbf{x})) \mathbf{g}_1(\xi) \mathrm{d}V_{\mathbf{x}'} \right\|}, & \text{if } \|\nabla s\| > 1.0 \times 10^{-2} \Delta x \\ 0, & \text{else} \end{cases}$$

$$\kappa^*(\mathbf{x}) = -\nabla \cdot \mathbf{n}(\mathbf{x}) = -\int_{H_{\mathbf{x}}} \left(\mathbf{n}(\mathbf{x}') - \mathbf{n}(\mathbf{x}) \right) \cdot \mathbf{g}_1(\xi) \mathrm{d}V_{\mathbf{x}'}$$

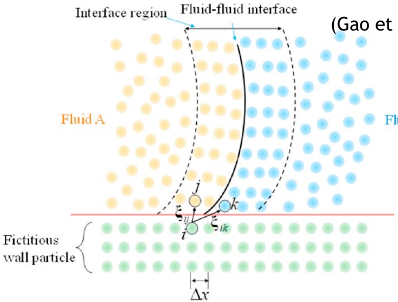
Introduce correction factors to correct the curvature field of boundary particles

$$\text{let } \mathbb{N}(\mathbf{x}) = \begin{cases} 1, & \text{if } \|\nabla s\| > 1.0 \times 10^{-2} \Delta x \\ 0, & \text{else} \end{cases}$$

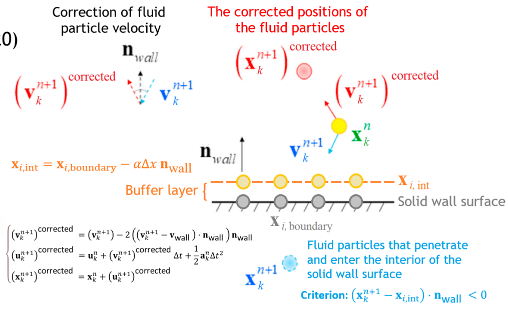
$$\varsigma^*(\mathbf{x}) = \frac{\int_{H_{\mathbf{x}}} \min\left(\mathbb{N}(\mathbf{x}), \mathbb{N}(\mathbf{x}') \right) w_0(\|\boldsymbol{\xi}\|) \mathrm{d}V_{\mathbf{x}'}}{\int_{H_{\mathbf{x}}} w_0(\|\boldsymbol{\xi}\|) \mathrm{d}V_{\mathbf{x}'}}, \kappa^{**}(\mathbf{x}) = \frac{\kappa^*(\mathbf{x})}{\varsigma^*(\mathbf{x})}$$

(Gao et al., 2020)

Boundary handling



(Gao et al., 2020)



$$\begin{cases} \left(\mathbf{v}_k^{n+1}\right)^{\text{corrected}} = \left(\mathbf{v}_k^{n+1}\right) - 2\left(\left(\mathbf{v}_k^{n+1} - \mathbf{v}_{\text{wall}}\right)\cdot \mathbf{n}_{\text{wall}}\right)\mathbf{n}_{\text{wall}} \\ \left(\mathbf{u}_k^{n+1}\right)^{\text{corrected}} = \mathbf{u}_k^{n} + \left(\mathbf{v}_k^{n+1}\right)^{\text{corrected}}\Delta t + \frac{1}{2}\mathbf{a}_k^n \Delta t^2 \\ \left(\mathbf{x}_k^{n+1}\right)^{\text{corrected}} = \mathbf{x}_k^{n} + \left(\mathbf{u}_k^{n+1}\right)^{\text{corrected}} \end{cases}$$

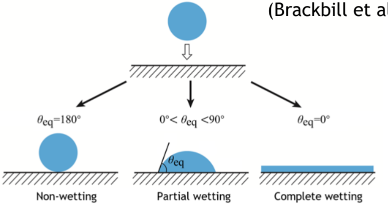
Considering the no-slip boundary, the fluid velocity is zero at the wall surface.

$$\mathbf{v}_i^n = 2\mathbf{v}_{\text{wall}} - \sum_{k=1}^{N_f} w_0\left(\left\|\boldsymbol{\xi}_{ij}^n\right\|\right)\mathbf{v}_k^n$$

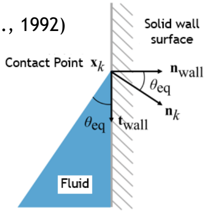
$$P_i = \frac{\sum_{k=1}^{N_f}\left(P_k + \left(\mathbf{F}_k^B - \rho_k \mathbf{a}_{\text{wall}}\right)\cdot \boldsymbol{\xi}_{ik}^n\right) w_0\left(\left\|\boldsymbol{\xi}_{ij}^n\right\|\right)}{\sum_{k=1}^{N_f} w_0\left(\left\|\boldsymbol{\xi}_{ij}^n\right\|\right)}$$

$$\rho_i = \rho_{0,i}\left(\frac{P_i - P_{0,i}}{P_{ref,i}} + 1\right)^{1/\gamma_i}$$

Consider the surface properties of the solid



(Brackbill et al., 1992)



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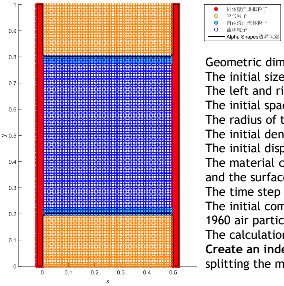
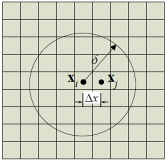
Numerical example

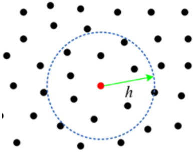


Geometric dimensions: 0.5 cm × 1.0 cm.
The initial size of the central liquid bridge is 0.5 cm × 0.6 cm.
The left and right boundaries are stationary rigid solid walls.
The initial spacing between the material points Δx = 0.01 cm.
The radius of the near-field domain δ = 3.015 Δx.
The initial density of the fluid particles is set to ρ_0 = 1 kg/m³.
The initial displacement, velocity, acceleration and pressure are all zero.
The material constant of the fluid γ = 1, the artificial sound speed c_f = 12 cm/s, the dynamic viscosity coefficient μ = 0.2 Ps·s, and the surface tension coefficient ß = 1 N/m.
The time step $\Delta t = 5.0 \times 10^{-7}$ s, and the total number of time steps N = 15000.
The initial computational domain was discretized into 2989 fluid particles, 606 fictitious particles representing solid walls, and 1960 air particles.
The calculation program for the example was written using **MATLAB 2015b.**
Create an independent main program file that encompasses all the calculation processes, rather than adopting the method of splitting the main program into multiple sub-program scripts and then calling them.

GitHub https://github.com/Liskelleo/Numerical-Simulation-of-the-Evolution-of-Water-Meniscus-Based-on-Peridynamics

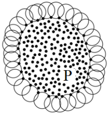


Two-dimensional structure with central orthogonal uniform discretization.

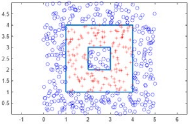


By calling the library function rangesearch to conduct a KD-tree neighborhood search for the specified particle.

The free surface is identified by calling library functions such as:



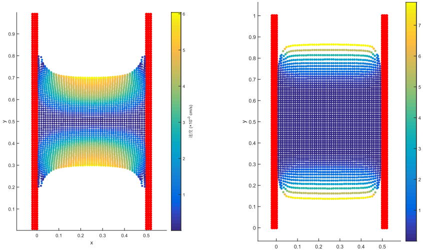
alphaShape



inpolygon

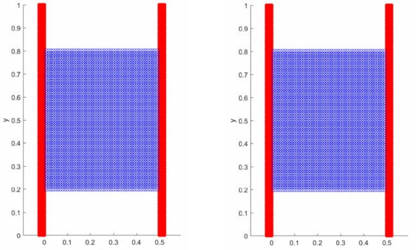
Results

Final configuration



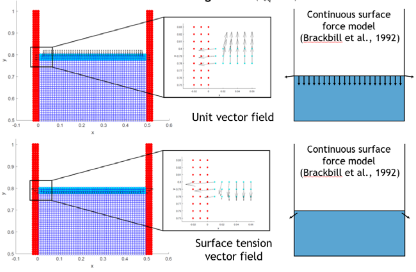
(a) $\theta_{eq} = 5°$ (b) $\theta_{eq} = 175°$

Liquid level evolution process

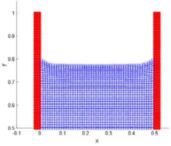


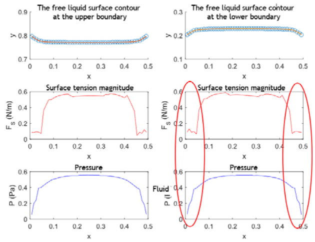
(a) $\theta_{eq} = 5°$ (b) $\theta_{eq} = 175°$

Initial configuration ($\theta_{eq} = 5°$)



Particle distribution at 7500 time steps for the concave liquid surface example ($\theta_{eq} = 5°$)

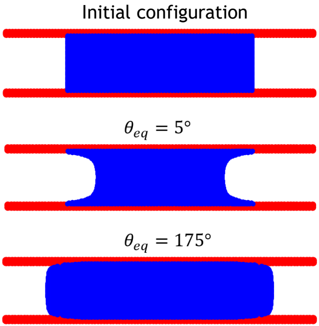


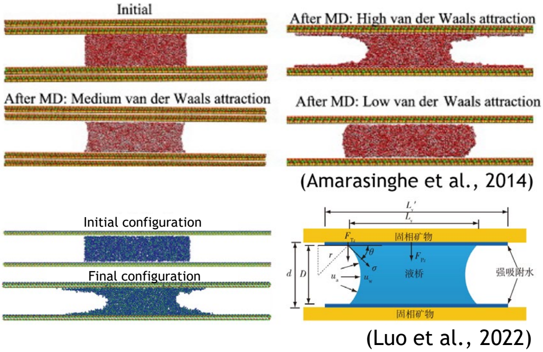


Note that the absolute value of the component of surface tension along the y-axis is selected to replace the surface tension for analysis.

PD vs. MD

The stronger the adsorption of the solid boundary on the pore water, the more "concave" the curved liquid surface will be; conversely, it will be more "convex".





(a) PD results

(b) MD results

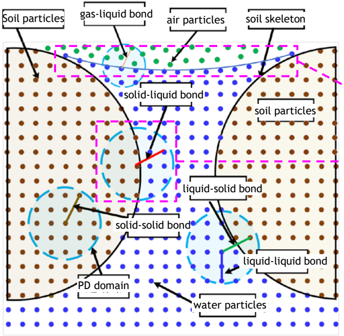
Contents

Future Work

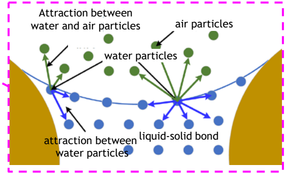
Simplified single liquid bridge model

Capillary action ✓
Evaporation effect ✗

Investigation on the Evolution of Curved Liquid Surface under the Coupling Effect of Heat, Water and Force is our future work!



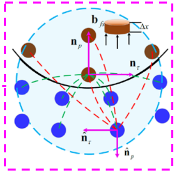
The microscopic structure of clay



One-way gas-liquid interaction

Boundary conditions:

Gas → Liquid: Surface tension
Solid ⇄ Liquid: Decomposes into two forces acting in the pressure direction and the shear direction



Fluid-solid interaction

Thanks for your time and attention!